KAGURA PICO CARRIER / BASIC REVA

72 x 48 mm, 4 layers / top Tang + XIAO, bottom Pico
44 PCB components; 28 top-side SMT placements at JLCPCB
Named global nets connect all four functional sheets.

Power

14 components / J3, F1, SW1, Q1, C5, C6, R6, R7, U5, D4, R12, D1, D2, D3

ファイル: Power.kicad_sch

USB Host

14 components / U2, J2, U4, C4, C1, C3, C8, R3, R11, JP1, R4, R5, R1, R2

ファイル: USB_Host.kicad_sch

Modules

10 components / J101, J102, U1, J2012, J2011, J2013, J2014, J2021, J2022, J2023

ファイル: Modules.kicad_sch

Expansion

6 components / J4, J5, C7, R10, R8, R9

ファイル: Expansion.kicad_sch

PCB-synchronized / physical socket contacts / prototype review pending

Sheet: /
File: KAGURA_PicoCarrier_RevA.kicad_sch

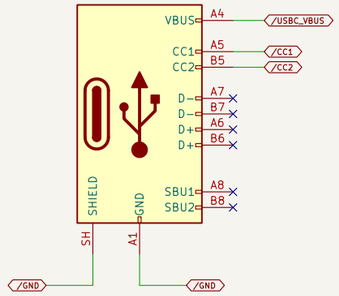
Title: KAGURA Pico Carrier — system overview

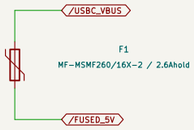
Size: A4	Date: 2026-11-12	**Rev: Basic RevA**
KiCad E.D.A. 10.0.8		Id: 1/5

POWER INPUT, SWITCH AND ISOLATED MODULE RAILS

5V input only. SW1 changes the PMOS gate; NC tabs are mechanical.
Diodes: pin 1 = cathode. Startup, thermal and reverse leakage qualification pending.

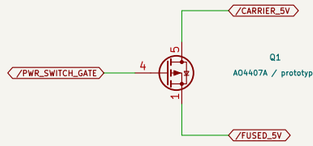
J3
USB4105-GF-A / 5V3A POWER ONLY

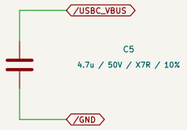


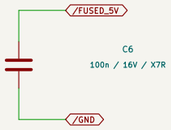


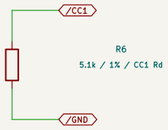
SW1
SK-12D02-G040 / C2930976

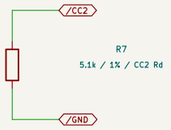


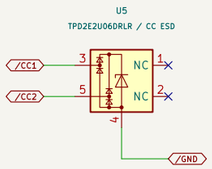


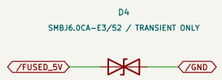


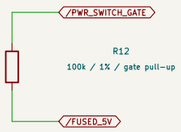






















PCB-synchronized / physical socket contacts / prototype review pending

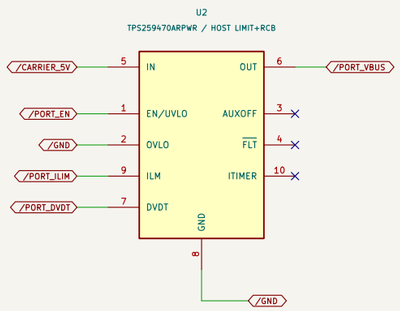
Sheet: /Power/
File: Power.kicad_sch

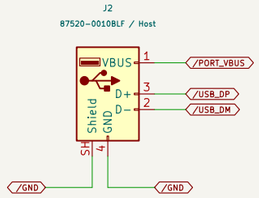
Title: Power input, switch and isolated module rails

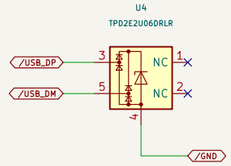
Size: A3	Date: 2026-11-12	**Rev: Basic RevA**
KiCad E.D.A. 10.0.8		Id: 2/5

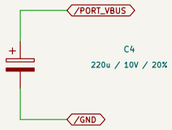
USB HOST, EFUSE AND DATA PROTECTION

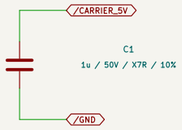
TPS259470A limits and isolates host VBUS. JP1 open = host power OFF.
Q1 drive, C4/U2 land patterns, inrush and voltage margins remain prototype checks.

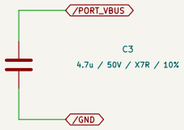


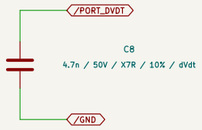


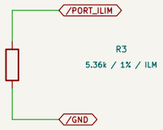


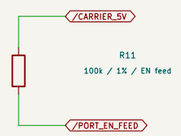


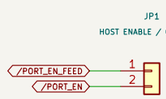


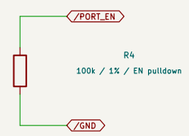


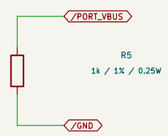


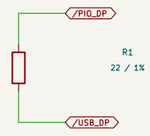


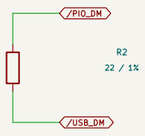












PCB-synchronized / physical socket contacts / prototype review pending

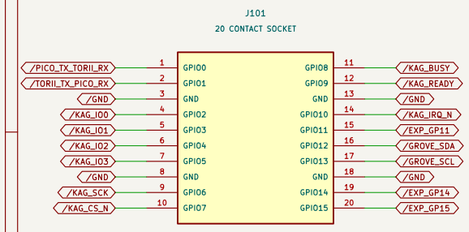
Sheet: /USB Host/
File: USB_Host.kicad_sch

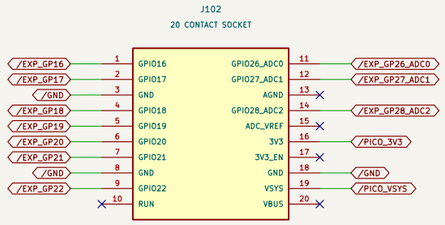
Title: USB host, eFuse and data protection

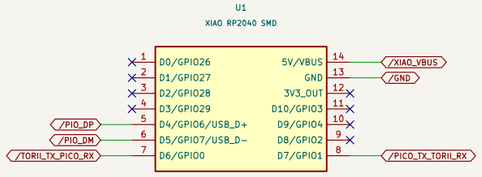
Size: A3	Date: 2026-11-12	**Rev: Basic RevA**
KiCad E.D.A. 10.0.8		Id: 3/5

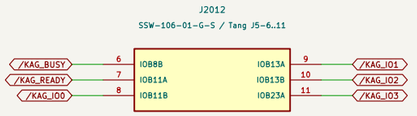
PICO SOCKETS, TANG SPLIT SOCKETS AND XIAO

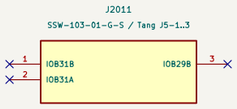
Physical carrier socket numbering, not an additional copy of the module circuit.
Tang support contacts marked NC remain physically populated. Pico is on the PCB bottom.

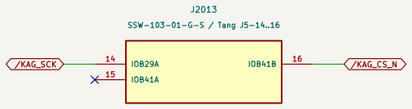


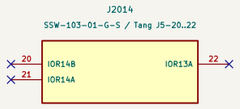


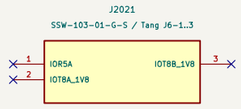




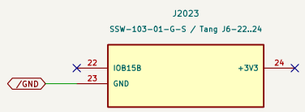












PCB-synchronized / physical socket contacts / prototype review pending

Sheet: /Modules/
File: Modules.kicad_sch

Title: Pico sockets, Tang split sockets and XIAO

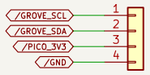
Size: A3	Date: 2026-11-12	**Rev: Basic RevA**
KiCad E.D.A. 10.0.8		Id: 4/5

GROVE I2C, GPIO EXPANSION AND IRQ BIAS

Grove is 3.3V only; shared external load budget 50mA.
J5 is DNP. Module-side pin headers and complete modules are separate purchases.

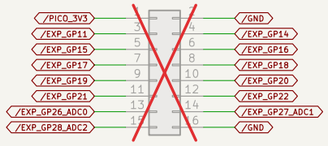
J4

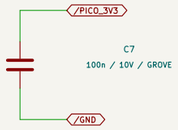
GROVE I2C / 3V3 ONLY / 50mA SHARED



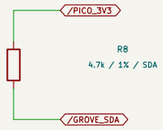
J5

GPIO EXPANSION / 3V3 ONLY











PCB-synchronized / physical socket contacts / prototype review pending

Sheet: /Expansion/
File: Expansion.kicad_sch

Title: Grove I2C, GPIO expansion and IRQ bias

Size: A3	Date: 2026-11-12	**Rev: Basic RevA**
KiCad E.D.A. 10.0.8		Id: 5/5